GGG–V–415a

April 4, 1967

SUPERSEDING
Int. Fed. Spec. GGG–V–00415(GSA–FSS)
March 31, 1965 and
Fed. Spec. GGG–V–436d
June 20, 1958 (in part) (See 6.3)

FEDERAL SPECIFICATION

VISE, PIPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vises for holding various sizes of pipe and are suitable for pipe stand mounting.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Pipe vises covered by this specification shall be of one type and the following classes as specified (see 6.1):

Type I.—Pipe.
- Class 1.—Chain.
- Class 2.—Hinged jaw.
- Class 3.—Open side.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification:

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–H–15424—Hand Tools, Packaging of.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids, or request for proposal, shall apply.

National Bureau of Standards (NBS) Handbook:
H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive, and are not intended to preclude the purchase of vises otherwise conforming to this specification.

3.2 Materials.

3.2.1 Materials used in the construction of vises shall be as hereinafter specified. All castings shall be of uniform quality, free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Strength and other essential physical properties of the casting shall be adequate throughout for the purpose intended. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended.

3.2.2 Coatings. All unmachined surfaces of the component parts of the vise may have an oil coating; or shall be given either one coat of baked enamel or lacquer; or one coat of primer and one coat of air-drying enamel or lacquer. In no case shall fillers be used.

3.3 Identification marking. Each vise shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.4 Interchangeability of parts. All parts having the same manufacturer's number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.5 Vise screw. The vise screw shall be made from steel of not less than 55,000 pounds per square inch (p.s.i.) yield and 65,000 p.s.i. tensile strength (see 4.4.2).

3.5.1 Screw threads. The screw threads used on vises shall be in accordance with NBS H28.

3.6 Type I, pipe vise.

3.6.1 Class 1, chain. Class 1 vise shall consist of a base, jaw(s), screw, chain, handle, and nut.

3.6.1.1 Base. The base shall be of forged steel, malleable iron, or ductile cast iron and have at least three mounting holes. The base shall be notched or other means shall be provided to anchor the chain by means of the chain pins.

3.6.1.2 Jaw(s). The jaw(s) shall be of steel, ductible iron, or flame-hardened pearlitic malleable iron, and either rigidly mounted or forged integral with the base. The clamping surface of the jaw(s) shall be V-shaped or semicircular and have mill cut V-shaped teeth for gripping the pipe. The jaw(s) shall show a Rockwell hardness value of not less than 52 nor more than 60 on the C scale (see 4.4.1).

3.6.1.3 Chain. The chain shall be of steel having a tensile strength of not less than 90,000 p.s.i. (see 4.4.2), and be of the flat link type with projecting link pins to engage the slot in the base. The chain shall be replaceable and of such length as to grip the pipe of the maximum size for which the vise is designed.

3.6.1.4 Handle. The handle shall be of forged steel, malleable iron, ductile iron, cast iron, or integral pin and yoke section of forged steel and secured to the nut.

3.6.1.5 Nut. The nut shall be malleable iron, ductile iron or steel, threaded internally to engage the thread of the screw. The length of the nut shall be at least equal to the major diameter of the screw.

3.6.1.6 Class 1 vise shall conform to table I for the size specified (see 6.1), and shall be similar to figure 1.

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Table I. Type I, class 1, pipe, chain.

* Nominal pipe size capacity	Screw diameter (minimum)	Weight (minimum)
Inches	Inch	Pounds
1/8 to 2 1/2	11/16	3 3/4
1/4 to 4	3/4	10
3/8 to 6	3/4	18
1/2 to 8	3/4	30

* Capacities noted are minimum and maximum required but are not intended to preclude vises with capacity below the minimum or above the maximum (by not more than 1 inch), provided all other requirements are met.

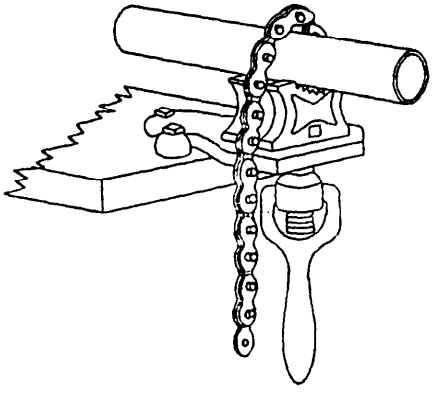


Figure 1. Type I, class 1, pipe, chain.

3.6.2 Class 2, hinged jaw. The vise shall consist of a base, frame and hook, jaw crosshead, jaws, screw, and screw handle.

3.6.2.1 Base. The base shall be of malleable iron, ductile iron, or steel and have at least three mounting holes. The vertical upright section of the base shall be provided with holes for the mounting of the frame, and shall have provisions on both sides for automatic engagement of the latch hook. The upright section shall be symmetrical about the vertical centerline to provide for hinging the frame from either the right or left side. The lower jaw mounting shall be so designed as to allow the vise to clamp the minimum pipe-size capacity of the vise.

3.6.2.2 Frame and hook. The frame and hook shall be of malleable iron, ductile iron, or steel and shall be U-shaped. The open ends of the frame shall have mounting boltholes to align with the boltholes in the uprights of the base. The closed end shall have provision for an internal threaded hole in the center of the frame or a means to hold a replaceable nut to engage the vise screw. One of the legs shall have a hole for mounting the latch hook; the latch hook shall be of steel or malleable iron and shall be designed and mounted to automatically lock the frame in the closed position. The assembled frame shall be so designed as to be reversible in a vertical plane to the base to provide for hinging the frame either from the right or left side of the base. The inside edge of the frame shall have smooth and parallel ways free of burrs and fins to act as guides for the jaw crosshead and shall permit the crosshead to slide freely over the entire length of adjustment without binding.

3.6.2.3 Jaw crosshead. The jaw crosshead shall be of malleable iron, ductile iron, or steel. The design shall be such that the strength will be adequate for the purpose intended. The top of the crosshead shall provide means to allow the end of the screw to swivel and be designed to both raise and lower the crosshead by action of the screw.

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3.6.2.4 Jaws. The jaws shall be of steel heat treated to a Rockwell hardness value of not less than 52 nor more than 60 on the C scale (see 4.4.1). The gripping surface shall be V or semicircular in shape and have mill cut V-shaped teeth. The lower jaw mounting shall be designed to allow the vise to clamp the minimum pipe-size capacity of the vise.

3.6.2.5 Screw. The vise screw shall have American National, square, acme, modified acme, or buttress threads and be in accordance with 3.5. One end of the screw shall have a steel head of adequate size which shall be either an integral part of the screw or welded to the screw in a permanent manner. The head shall be provided with a hole to receive the screw handle. The screw shall engage the full length of the threaded bearing of the body when the jaw is at the maximum opening indicated in the applicable table.

3.6.2.6 Screw handle. The screw handle shall be a straight, round steel bar of a diameter comparable to the size of the vise. The screw handle shall slide freely in the hole of the screwhead with means provided at each end of the handle to pervent complete disengagement from the screwhead.

3.6.2.7 Class 2 vise shall conform to table II for the size specified (see 6.1), and shall be similar to figure 2.

TABLE II. Type I, class 2, pipe, hinged jaw.

* Nominal pipe size capacity	Screw diameter (minimum)	Weight (minimum)
Inches	Inches	Pounds
1/8 to 2 1/2	7/8	10
1/8 to 4 1/2	1	23
1/8 to 6	1 1/8	35
1 to 8	1 1/8	75
1 1/2 to 12	1 3/8	160

* Capacities noted are minimum and maximum required but are not intended to preclude vises with capacity below the minimum or above the maximum (by not more than 1 inch), provided all other requirements are met.

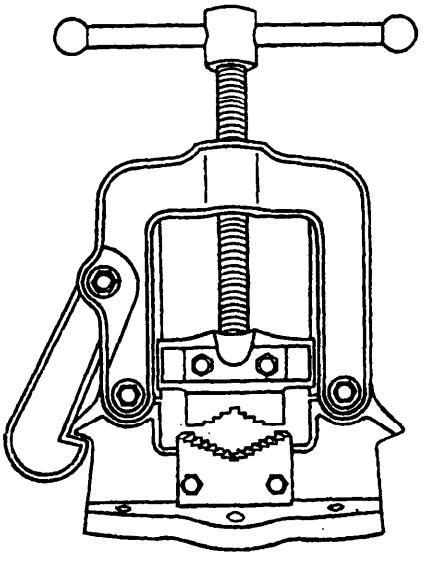


Figure 2. Type 1, class 2, pipe, hinged jaw.

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3.6.3 Class 3, open side. Class 3 vise shall consist of a base with column, jaw slide, jaws, screw, and screw handle.

3.6.3.1 Base with column. The base shall be of malleable iron or steel with at least three mounting holes. The columns shall be an integral part of the base and have smooth and parallel guide ways free of burrs and fins or a guide bar exclusive of the column for the upper jaw slide, and have a threaded hole in the top portion of the column to engage the vise screw.

3.6.3.2 Jaw slide. The jaw slide shall be of malleable iron or steel with provision for mounting the upper jaw and shall be guided through the entire capacity of the column without binding. One end of the screw shall swivel in the top section of the slide and the attachment shall be designed to allow for both raising and lowering the upper jaw by means of the vise screw.

3.6.3.3 Jaws. The jaws shall be of steel and conform in detail and method of mounting as specified for class 2 vise (see 3.6.2.4).

3.6.3.4 Screw. The screw shall conform to the requirements specified in 3.6.2.5.

3.6.3.5 Screw handle. The screw handle shall conform to the requirements specified in 3.6.2.6.

3.6.3.6 Class 3 vise shall conform to table III for the size specified (see 6.1), and shall be similar to figure 3.

TABLE III. Type I, class 3, pipe, open side.

* Nominal pipe size capacity	Screw diameter (minimum)	Weight (minimum)
Inches	Inches	Pounds
1/8 to 1 1/2	3/4	6 1/2
1/8 to 2 1/2	7/8	11 1/2
1/8 to 4 1/2	1 1/8	27

* Capacities noted are minimum and maximum required but are not intended to preclude vises with capacity below the minimum or above the maximum (by not more than 1 inch), provided all other requirements are met.

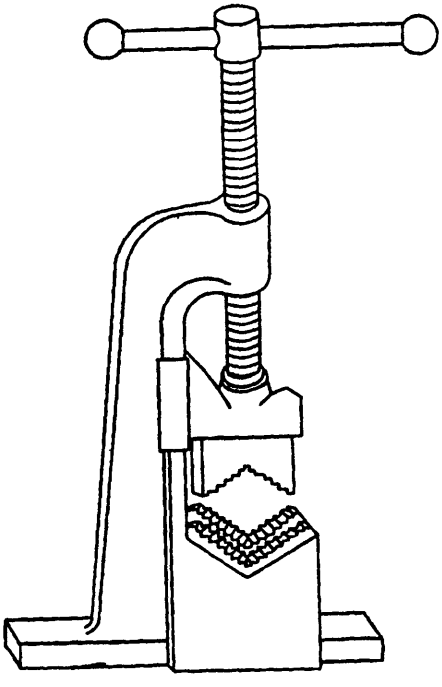


Figure 3. Type I, class 3, open side.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table IV.

TABLE IV. Sampling data.

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	4.0	Defects per hundred units	4.3.1
Dimensional and weight examination	do.	S-2	4.0	do.	4.3.2
Testing	do.	S-2	2.5	do.	4.4.1
	do.	S-1	4.0	do.	4.4.2
Preparation for delivery	1 container	S-2	4.0	do.	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table V.

TABLE V. Classification of defects.

Examine	Defects
Material	Not fabricated of specified material.
Coating	Missing or not as specified, not smooth, uniform or adherent.
Marking	Missing, incorrect, illegible, or not permanent.
Design	Any characteristic not in accordance with specified requirements.
Construction and workmanship	Incomplete, parts missing, not fitted and assembled as required; provisions not made to prevent parts becoming detached while in use.
	Moving parts bind or are excessively loose.
	Threads nonconforming, crossed, stripped, chipped or torn.
	Not smoothly finished; not free from burrs, fins or cracks.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 Hardness test. Hardness test shall be conducted in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine compliance with 3.6.1.2 and 3.6.2.4.

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4.4.2 Tension test. Each sample vise for which the applicable physical requirements of the steel are specified in 3.5 and 3.6.1.3 shall be tested in accordance with method 211 of Fed. Test Method Std. No. 151.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VI. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL–STD–105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery.

TABLE VI. Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of containers.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified (see 6.1), the vises shall be preserved, packaged and packed in accordance with MIL–H–15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.1).

5.1.1 As part of the requirements of section 5 of this specification the following is considered as part of table I of MIL–H–15424.

COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5	COLUMN 6
Vise:					
Pipe:					
Chain					
Hinged jaw					
Open side					
up through 5 pounds	P–2, P–5	I	1		24
6 through 10 pounds	P–2, P–5	I	1		12
11 through 20 pounds	P–2, P–5	I	1		8
21 through 50 pounds	P–2, P–5	I	1		2
51 pounds and over	P–2, P–5	I	1		1

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein, and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Class of vise required (see 1.2).
(c) Size required (see applicable tables).
(d) Selection of levels of preservation, packaging, and packing required (see 5.1 and 5.2).

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6.1.1 When replacement parts are required, the procurement document should specify the following:

(a) The name or trademark of the manufacturer of the vise for which the part is required.
(b) Proper name or number of the part required.
(c) Quantity of each item required.
(d) Other data or description applicable to the vise which will assist in identifying the proper replacement parts.

6.2 For civil agency procurements. When level B packaging is required for civil agency procurements, the requirements in 5.1.1, level A packaging shall apply.

6.3 Supersession data. The classification of vises covered by this specification corresponding to those covered in GGG–V–436d are as follows:

GGG–V–436d	GGG–V–415a
Type X.—Pipe.	Type I.—Pipe.
Class 1.—Chain.	Class 1.—Chain.
Class 2.—Hinged jaw.	Class 2.—Hinged jaw.
Class 3.—Open side.	Class 3.—Open side.

MILITARY INTEREST:

Army—GL	Future coordination waived.
Navy—SH	Coordination waived.
Air Force—69	Coordination waived.

Civil Interest:

GSA–FSS

Preparing Activity:

GSA–FSS

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